3.6.1 Stimuli, both internal and external, are detected and lead to a response (A-level only)

3.6.1.1 Survival and response (A-level only)

Content	Opportunities for skills development
Organisms increase their chance of survival by responding to changes in their environment. In flowering plants, specific growth factors move from growing regions to other tissues, where they regulate growth in response to directional stimuli. The effect of different concentrations of indoleacetic acid (IAA) on cell elongation in the roots and shoots of flowering plants as an explanation of gravitropism and phototropism in flowering plants. Taxes and kineses as simple responses that can maintain a mobile organism in a favourable environment. The protective effect of a simple reflex, exemplified by a three-neurone simple reflex. Details of spinal cord and dorsal and ventral roots are **not** required.	**AT h** Students could design and carry out investigations into the effects of indoleacetic acid on root growth in seedlings.
Required practical 10: Investigation into the effect of an environmental variable on the movement of an animal using either a choice chamber or a maze.	**AT h**

3.6.1.2 Receptors (A-level only)

Content	Opportunities for skills development
The Pacinian corpuscle should be used as an example of a receptor to illustrate that: • receptors respond only to specific stimuli • stimulation of a receptor leads to the establishment of a generator potential. The basic structure of a Pacinian corpuscle. Deformation of stretch-mediated sodium ion channels in a Pacinian corpuscle leads to the establishment of a generator potential. The human retina in sufficient detail to show how differences in sensitivity to light, sensitivity to colour and visual acuity are explained by differences in the optical pigments of rods and cones and the connections rods and cones make in the optic nerve.	**AT h** Students could design and carry out investigations into: • the sensitivity of temperature receptors in human skin • habituation of touch receptors in human skin • resolution of touch receptors in human skin.

3.6.1.3 Control of heart rate (A-level only)

Content	Opportunities for skills development
Myogenic stimulation of the heart and transmission of a subsequent wave of electrical activity. The roles of the sinoatrial node (SAN), atrioventricular node (AVN) and Purkyne tissue in the bundle of His. The roles and locations of chemoreceptors and pressure receptors and the roles of the autonomic nervous system and effectors in controlling heart rate.	**AT h** Students could design and carry out an investigation into the effect of a named variable on human pulse rate. **MS 2.2** Students could use values of heart rate (*R*) and stroke volume (*V*) to calculate cardiac output (*CO*), using the formula $CO = R \times V$